

ChargeCascade: A Tumbling-Mill Charge-Motion and Power Workbench, and the Honest Disagreement of Standard Power Models

Felipe Santibañez-Leal 

Abstract—The power a tumbling mill (SAG, ball or rod) draws is the single largest electricity cost in a concentrator, and predicting it, and the charge motion behind it, is a classic problem with more than one accepted answer. ChargeCascade is an interactive workbench that computes tumbling-mill charge motion and power live in the browser from geometry, fill, ball size and the fraction of critical speed, and, unusually, it shows where the accepted models *disagree* rather than hiding it behind one. The charge motion follows Davis (1919) kinematics: a per-shell departure angle traces the shoulder, toe and cataract fan across the cascading, cataracting and centrifuging regimes, with the centrifuging onset landing exactly at the critical speed. The power is computed by two standard closed-form models, Hogg-Fuerstenau (with a disclosed effective-lift-arm calibration) and Morrell, and the report’s central honest finding is that these two textbook models *disagree by about 23%* (median, up to 29%) across ten cases, with a third estimate from a baked discrete-element simulation (5000–10000 particles) landing around them rather than adjudicating between them. This is the true state of mill-power prediction, real cross-method uncertainty of order twenty percent, and the workbench displays all three side by side rather than pretending to a false precision. A learned surrogate reproduces the closed-form power to 5.2% for instant what-ifs, guarded by an out-of-distribution detector that flags settings outside its training envelope. The engine is a single dependency-free source of physics truth; every number and figure regenerates from the committed artifacts.

Index Terms—Tumbling mill, SAG mill, ball mill, charge motion, mill power, Davis kinematics, Hogg-Fuerstenau, Morrell, discrete element method, surrogate model, comminution.

I. INTRODUCTION

A tumbling mill grinds ore by lifting a charge of balls and rock on a rotating shell and letting it cascade and cataract back down, and the power it draws to do so is the dominant energy cost of a concentrator. Predicting that power, and the charge motion that produces it, has a century of history and, importantly, more than one accepted model. The charge kinematics trace back to Davis [1]: at a fraction $\phi_c = N/N_c$ of the critical speed $N_c = 42.3/\sqrt{D-d}$, a ball on a shell of radius r leaves the shell at a departure angle given by $\cos \alpha = \phi_c^2 r/R$ and follows a ballistic arc, and summing over shells gives the shoulder, the toe and the cataract fan. The power has at least two standard closed-form answers, the Hogg-Fuerstenau torque-arm model [2] and the Morrell

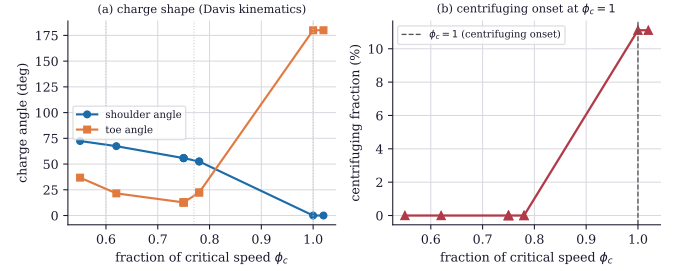


Fig. 1. Charge motion by Davis kinematics, across ten cases. (a) The shoulder and toe angles against the fraction of critical speed ϕ_c : as ϕ_c rises the charge lifts higher (the shoulder rises, the toe advances) through the cascading and cataracting regimes, and at the critical speed the charge centrifuges (the shoulder collapses to zero and the toe to 180°, the charge pinned to the shell). (b) The centrifuging fraction is zero below the critical speed and turns on exactly at $\phi_c = 1$, the exact onset Davis kinematics predicts.

model [3], and a higher-fidelity but expensive answer from the discrete element method [6], [7], [8]. These do not all agree, and much software and teaching quietly picks one and reports it as *the* power.

ChargeCascade is a workbench built on the opposite principle: compute the charge motion and the power from the cited models, live and in the browser, and show the reader where the models disagree. It runs Davis kinematics for the charge shape and regime, both the Hogg-Fuerstenau and the Morrell closed-form power models, and a set of baked discrete-element traces as a third estimate, and it puts them side by side. The report describes the workbench and states its central honest finding: the two standard closed-form power models disagree by about a fifth, which is the real uncertainty of mill-power prediction, and displaying it is more useful than concealing it. A learned surrogate accelerates the closed-form engine, and an out-of-distribution guard flags extrapolation.

II. CHARGE MOTION: DAVIS KINEMATICS

The charge shape and regime follow from Davis kinematics evaluated over nine radial shells (Fig. 1). As the fraction of critical speed rises, the charge is carried higher before it departs, so the shoulder angle rises and the toe advances, and the motion passes through the literature regime bands: slumping and cascading at low speed (here $\phi_c \approx 0.55$ – 0.62 , shoulder near 70°), cataracting at intermediate speed ($\phi_c \approx 0.75$ – 0.78 , the ballistic fan that does the impact grinding in a SAG mill), and centrifuging at the critical speed. The centrifuging onset is exact: below $\phi_c = 1$ no shell centrifuges,

F. Santibañez-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanez@gmail.com; ORCID 0000-0002-0150-3246).

Technical report, 2026. Interactive workbench and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChargeCascade; live at <https://chargecascade.fasl-work.com>.

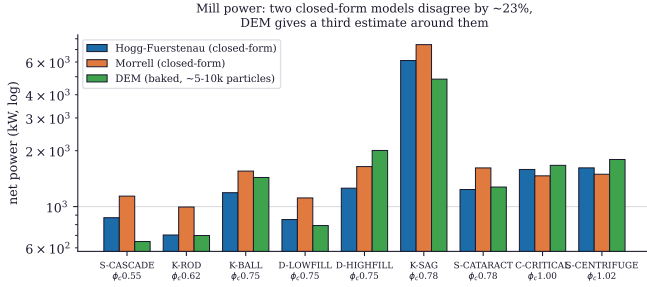


Fig. 2. Net mill power per case, on a log axis, from two closed-form models and from a baked discrete-element simulation. The Hogg-Fuerstenau and Morrell closed-form models *disagree by about 23%* (median; up to 29%); Morrell is consistently higher at cascading and cataracting speeds, and the two cross near the critical speed. The discrete-element estimate (5000–10000 particles) lands around them, sometimes below both, sometimes above, rather than settling the disagreement. This spread is the true uncertainty of mill-power prediction, and the workbench shows all three rather than reporting one as fact.

and at $\phi_c = 1$ the outer shell pins to the wall, the shoulder collapses to zero and the toe to 180° (Fig. 1b). Because the critical speed and ϕ_c are derived from geometry, not fitted, the regime map is a property of the kinematics, and it reproduces the textbook picture, which is the check that licenses using the workbench to reason about charge motion.

III. POWER: WHERE THE STANDARD MODELS DISAGREE

The power is where the honesty of the workbench matters most (Fig. 2, Table I). Two standard closed-form models are computed: Hogg-Fuerstenau [2], a torque-arm model with a disclosed effective-lift-arm calibration ($C_{\text{arm}} = 0.80$, stated in the interface), and Morrell [3]. They do not agree. Across the ten cases the two disagree by a median of 23% and by as much as 29%: Morrell draws consistently higher at the cascading and cataracting speeds that mills actually run at (for example 1139 kW against 872 kW on a cascading case), and the two only cross near the critical speed. A third estimate, from a baked discrete-element simulation of five to ten thousand particles, lands *around* the two closed-form values, below both on some cases and above on others, rather than adjudicating between them. This is not a defect of the workbench; it is the real state of the field, mill-power models carry cross-method uncertainty of order twenty percent, and a tool that reports a single number to four significant figures is asserting a precision that does not exist. ChargeCascade shows the two closed-form models and the discrete-element estimate together, with the Hogg-Fuerstenau calibration disclosed, so a user sees the spread and can reason about it.

IV. THE LEARNED SURROGATE AND ITS GUARD

For instant what-if exploration the workbench runs a learned surrogate of the closed-form engine: a small multilayer perceptron (six inputs, two hidden layers of sixty-four units, mapping to power and centrifuging fraction) trained on the engine’s outputs. It reproduces the closed-form power to a mean relative error of 5.2% (with the centrifuging fraction exactly, since it is a kinematic threshold), fast enough to

TABLE I
NET POWER BY THE TWO CLOSED-FORM MODELS AND THE BAKED DEM, REPRESENTATIVE CASES. THE CLOSED-FORM MODELS DISAGREE BY ~20–30%; DEM LANDS AROUND THEM. C-EMPTY (EMPTY MILL) IS A ZERO-POWER SANITY CHECK.

Case	ϕ_c	Hogg-F. (kW)	Morrell (kW)	DEM (kW)
S-CASCADE	0.55	872	1139	649
K-ROD	0.62	704	995	699
K-BALL	0.75	1188	1553	1435
D-HIGHFILL	0.75	1256	1642	2005
K-SAG	0.78	6111	7440	4852
C-CRITICAL	1.00	1584	1464	1667

update as a user drags a control. Crucially, because a surrogate is only trustworthy inside the envelope it was trained on, a companion out-of-distribution detector flags settings outside that envelope, so a user who pushes the sliders to an untrained corner is warned that the surrogate is extrapolating rather than being silently misled. A separate learned-DEM increment, a graph-network simulator trained to roll the particle dynamics forward, is under development as the bridge to the baked DEM lane; it is disclosed as in-progress, not presented as validated. The surrogate accelerates the engine; it does not replace the disclosure of the underlying model disagreement, which remains front and centre.

V. DISCUSSION AND SCOPE

The workbench’s contribution is as much epistemic as computational: it computes tumbling-mill charge motion and power from the cited models and, by showing the Hogg-Fuerstenau and Morrell models and the discrete-element estimate together, it makes the real uncertainty of mill-power prediction visible instead of hiding it. The scope is stated plainly. The Hogg-Fuerstenau effective-lift-arm calibration is a disclosed constant, not a fit to a specific mill, and it is shown; the Morrell model is used as published. The cases are physically reasonable synthetic configurations, not a specific plant, so the numbers are the models’ behaviour, not a prediction for a given mill. The baked discrete-element traces are a higher-fidelity but still-modelled estimate at laboratory particle counts, not a measured power, and the learned-DEM graph-network surrogate is disclosed as in-development. The value is a transparent, reproducible workbench that treats mill-power prediction with the honesty its cross-method spread demands. None of this is hidden; the interface states the same on its methodology pages.

VI. CONCLUSION

ChargeCascade computes tumbling-mill charge motion by Davis kinematics, reproducing the cascading-ataracting-centrifuging progression with an exact centrifuging onset at the critical speed, and computes the power by two standard closed-form models plus a baked discrete-element estimate. Its central honest finding is that the two textbook power models disagree by about a fifth, and it displays that spread with the third estimate around it, rather than reporting one number as fact, because a fifth is the real uncertainty of mill-power prediction.

A learned surrogate reproduces the closed-form power to 5.2% with an out-of-distribution guard. The disclosure of the model disagreement, and of the one disclosed calibration, is the report’s discipline, and the transparent, reproducible workbench is its contribution.

APPENDIX A REPRODUCTION

The committed artifacts under `data/derived/` carry the ten cases’ engine outputs (`case-results.json`: ϕ_c , regime, shoulder and toe angles, centrifuging fraction, and both the Hogg-Fuerstenau and Morrell power per case), the learned-surrogate metrics (`cc-learned.json`) and the baked DEM summaries (`data/dem/`). The figure scripts under `manuscripts/mill-charge-power/figures/` read those artifacts. The dependency-free physics engine is the single source of truth for the App, the experiments table and the offline bake, so re-running reproduces Table I and Figs. 1–2.

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